DH^{2}A-KT: Leakage-Audited Knowledge Attribution for Knowledge Tracing

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Abstract

Graph-augmented knowledge tracing often reports a single AUC as if every constructed graph component contributed. This paper asks which leakage-audited structural and temporal messages survive a matched-twin ablation. DH^{2}A-KT constructs train-only auxiliary knowledge, audits hyperedge membership leakage, and credits a message only when $\Delta\text{val} \geq +0.002$ versus a matched twin on all five paired seeds. Adding a Linear $\log(1+\Delta t)$ *branch* meets that rule on both the observed-incidence and capacity-matched zero-incidence backbones (5/5; mean Δval+0.00255 on the zero-incidence twin). Architecture-matched controls that keep the extra maps but zero or misalign the gap remain positive but below the rule (0/5; mean Δval+0.00183 / +0.00144). We therefore credit the timing package under our convention, but do not isolate aligned gap input as a separately credited source. Zeroing only the question–KC incidence *message* does not meet the rule (0/5).

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On XES3G5M, under a repeat-target-masked KC-expanded protocol, the credited-minimal control reaches test-only AUC 0.8295±0.0001. Training-envelope-matched GIKT is a contextual benchmark (0.8258±0.0001), not a tuning-matched superiority claim.
Keywords: Knowledge tracing, knowledge attribution, data leakage, educational data mining

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1. Introduction

Knowledge tracing (KT) estimates a learner's latent mastery of knowledge components (KCs) from interaction history, supporting adaptive practice, intelligent tutoring, and curriculum recommendation. The field has progressed from probabilistic and sequence models [1] to graph-augmented models that propagate information over auxiliary concept graphs [2, 3], and, most recently, toward hypergraph representations that capture higher-order relations among knowledge components [4, 5, 6] and dynamic, temporally-evolving graph structure [7].

This paper builds on [8]'s train-only leakage protocol. DH^{2}A-KT is a leakage-audited *attribution* setting: construct, audit, then ask which knowledge predicts. The evaluation arm QKC-T is a GIKT-family LSTM with Q$\leftarrow$KC incidence and a linear Δt branch (Section 4.3); Zero+Δt is the credited-minimal control. Teacher, forum, and video members remain interfaces. Fig. 1 is the pipeline.

1.1. Motivation

Two concrete gaps motivate this work:

- **Higher-order structure is under-audited in graph-KT.** Hypergraph-KT models [4, 5, 6] group multiple knowledge components into a single hyperedge, yet published systems rarely audit whether those hyperedges were constructed without leakage, and few expose an interface for entities beyond KCs (hints, teachers, forums, video) when such logs become available.

- **Graph provenance is rarely audited.** [8] shows that pooling a knowledge-component graph from the full interaction log before a learner-based train/test split lets held-out information reach a graph-augmented KT model even when the sequence optimizer itself trains only on the training fold. [8]'s protocol audits this channel for *pairwise* prerequisite/similarity edges; published hyperedge-level audits remain scarce.

1.2. Research Questions

This work is motivated by a broader question: *which leakage-audited knowledge actually improves next-step KT?* We decompose this into two questions, plus entities this paper does not claim to have learned:

- **RQ1** (§5.5; Tables 8–9): Under a leakage audit, which structural knowledge meets a matched-twin gate on this backbone?

- **RQ2** (§6.2; Tables 15, 10): Does the timing-branch gain survive under history-only input, and is aligned Δt isolated from the extra maps?

An exploratory Critic-gated pilot over frozen scores is reported after the predictive results (Section 6.4); it is not a third research question and not a contribution. Teacher, forum/discussion, and video nodes are defined in the schema (Section 3) but remain data-blocked on every corpus we use (Section 7.3): this paper establishes an interface for those entities, not empirical evidence of how they improve KT. A graph-only encoder and threefold manipulation check remain available as a forced-reliance diagnostic (Appendix A); they are not RQ1's evaluation arm.

1.3. Contributions

- **C1 (construction and audit).** A procedure that groups [8]'s audited, acyclic prerequisite edges into *path-derived concept-group* chain hyperedges (unordered member sets along directed paths; Section 4.1) and names three leakage classes that a pairwise projection does not express (Section 4.2); session+Hint hyperedges on FoundationalASSIST [9] (Section 6.7). Discussion-thread and teacher-intervention types remain data-blocked (Section 7.3). Construction is not claimed as an AUC lift.

- **C2 (which knowledge predicts).** An operating rule (Δval $\geq +0.002$ vs. a matched twin on five paired seeds; an internal convention, not a public registry) plus a repeat-target-masked KC-expanded protocol. Adding a Linear $\log(1+\Delta t)$ *branch* meets the rule on the QKC-T backbone (5/5 seeds pass; Table 15) and on a capacity-matched zero-incidence twin (5/5; test-only AUC 0.8295±0.0001; Table 9). Those twins add maps as well as gap input. T-zero / T-misaligned on the same maps remain positive but below the val rule (0/5; Table 10). We credit the timing package, not aligned gap input as a separate source. Zeroing only the train-only incidence *message* does not meet the rule (0/5;

Table 1: What this paper claims, and what it does not infer. Evidence types: 5-seed matched twin; single-seed diagnostic; exploratory pilot.

Claim	Evidence	Not inferred
Linear Δt *package* is credited on this backbone	5-seed gate 5/5 vs. no-time twins, observed and zero incidence	Aligned gap input is separately credited
Aligned gap vs. T-zero / T-misaligned is positive but below the rule	5-seed val Δ +0.00183/+0.00144, 0/5	Aligned time has no value; or no matched control exists
Q–KC incidence message is not credited	5-seed capacity-matched 0/5	Question–KC relations are universally useless
Hyperedge membership diagnostic responds to contamination	Full-log leak $0 \rightarrow 1$	The whole audit taxonomy is validated ($\lvert\rho\rvert$/TBMR)
Table 7 baselines locate the recipe	Training-envelope GIKT; inherited AKT/simpleKT defaults	Tuning-matched superiority

Table 8). History-only LSTM Δt is 2/5 seeds pass and is not credited under the 5/5 rule. QKC-T is the observed-incidence evaluation arm (five-seed test-only 0.8295±0.0001; three-fold mean 0.8295±0.0004); Zero+Δt is the credited-minimal control. Table 7 is a contextual benchmark: GIKT shares a training envelope (not a matched tuning search) and is 0.8258±0.0001. Teacher grouping and per-concept exponential forgetting are 0/5 (Table 13). HypergraphConv, hint-item hyperedges, a full Q-matrix, and a Junyi DAG subsample remain single-seed diagnostics below the gate. Graph-only v2 mean AUC 0.752 versus GKT 0.834 mixes encoder, budget, and protocol; it is not an isolated architecture cost.

2. Related Work

2.1. Graph- and hypergraph-based knowledge tracing

Pairwise KT graphs [2, 3] and dynamic variants [7] propagate over edges; hypergraph KT [4, 5, 6] and heterogeneous node types [10, 11] still draw members from KC/exercise/learner sets. Among the hypergraph-KT systems we read [4, 5, 6, 10, 11], Teacher, Hint, Forum, and Video appear as interfaces here rather than as evaluated members. This paper's quantitative core is a GIKT-class LSTM with audited Q←KC incidence (Section 4.3); hypergraph operators are single-seed diagnostics in Table 13, not five-seed credit tests.

2.2. Forgetting and temporal effects in KT

Sequence KT models have long treated elapsed time as a forgetting cue. [12] extend DKT with multiple forgetting-related features (lag from the previous interaction and past trial counts). HawkesKT [13] models skill-to-skill temporal cross-effects with a Hawkes-process kernel (WSDM 2021). [14] encode *step* recency (how many KC-rows since a concept last appeared) with learnable Fourier features; that is not the wall-clock $\log(1+\Delta t)$ we credit. Our P4 arm is narrower: a per-concept exponential $\exp(-\mathrm{softplus}(\theta_c)\cdot \mathrm{gap})$ on the same v4 backbone as Linear $\log(1+\Delta t)$. That one form not meeting the gate does not rule out power-law decay, data-driven half-lives, or Hawkes kernels; we do not claim to have tested those alternatives.

2.3. LLM agents and explanation faithfulness

Closest in scope is [15], which aligns LLM mastery reports with implicit DLKT states and evaluates explanation faithfulness with KC-overlap, an independent-generation control, and paired ablations. Section 6.4 is an exploratory pilot in that line, not a third pillar: we gate rationales with a Critic over *frozen* P(correct) and use ID-anchored KC-Jaccard because XES3G5M has no in-repo synonym KC vocabulary. Broader LLM-agent surveys [16, 17, 18, 19] motivate the frozen-core split: re-inferring mastery with a multi-agent LLM over millions of interactions is not tractable on one consumer GPU.

2.4. Leakage-controlled evaluation

We searched ACM Digital Library, IEEE Xplore, and Scopus through September 2026 with the terms "hypergraph knowledge tracing", "leakage audit", "graph provenance", "KT label leakage", "question-level evaluation", and "controlled attribution". [8] (companion P0, under review) audits pairwise prerequisite/similarity edges under train-only construction. Within that search we found no published protocol that applies the same leakage diagnostics to *hyperedges*; this paper extends the audit from pairwise edges to hyperedges (Section 4.2). We reuse P0's train-only construction, four pairwise diagnostics, and pinned artefacts (Section 3.1). [14] (arXiv:2508.17092) targets a different leakage channel: KC-expansion *label* leakage, via a MASK token in the input embedding, plus Fourier recency (steps since a KC last appeared). That work does not audit hyperedge provenance and does not run capacity-matched attribution twins. Our clean protocol masks repeat-row *targets* at evaluation (pyKT-aligned) rather than replacing labels in the embedding; Linear $\log(1+\Delta t)$ is wall-clock attempt context, not step-count recency encoding.

Observational treatment-effect estimators in education [20, 21, 22] are background only.

3. Preliminaries

Let $\mathcal{D} = \{(u, t, q, c, y)\}$ be a KT interaction log with learner u, timestamp t, item q, knowledge component c, and correctness label $y \in \{0, 1\}$, following [8]'s notation. A learner-based temporal split $\mathcal{F} = (\mathcal{F}_{\text{train}}, \mathcal{F}_{\text{val}}, \mathcal{F}_{\text{test}})$ assigns each learner to exactly one fold (ratios 0.7/0.1/0.2, three folds with split seeds 42/43/44, matching [8]'s protocol so results remain directly comparable, see Section 5.1).

3.1. Self-contained summary of the reused P0 protocol

Because [8] is under review, we state the reused pieces here. For each learner-based fold, E^f_{pre} and E^f_{sim} are built from *train-fold interactions only*, then cycle-pruned to a DAG. Four pairwise diagnostics (P0 Table of leakage metrics; computed on the train-only graph vs. held-out learners) are:

- ECR^{flag}: structural learner-overlap indicator (whether any learner appears in more than one split);
- $\text{ECR}^{\text{overlap}}$: held-out pattern overlap on retained edges;
- $|\rho|$: magnitude of the Pearson correlation between edge weight and held-out outcome;
- TBMR (P0 also writes TBVR): *within-train* temporal mixing — the share of post-cutoff *training* mass that supports retained prerequisite transitions — *not* a train/test membership violation.

P0 reports that switching from full-log to train-only graph construction shifts baseline AUC by at most 0.003 on XES3G5M, ASSISTments 2012, and Junyi.

We pin the companion repository to commit `5e3d6a0` and copy its published baseline CSVs verbatim (Section 5.2). The companion PDF can be attached as journal supplementary material on request. Chain hyperedges in this paper are bounded-length paths on the same audited E^{f}_{pre} (Section 4.1).

We extend this notation with a hyperedge $h = (\kappa, \{(\tau_i, e_i)\}, f)$: kind $\kappa \in \{\text{path-derived concept-group, session, discussion-thread, teacher-intervention}\}$, typed members (student, exercise, concept, teacher, hint, forum, discussion, video), and fold f. Chain hyperedges on audited E_{pre} are audited and used in diagnostic arms; they do *not* enter QKC-T or Zero+Δt (Table 7). Teacher/forum/video kinds are interfaces; session write-backs appear only in the Tier 2 pilot. QKC-T consumes train-only Q←KC incidence and timestamps (Section 4.3); Zero+Δt keeps the timestamps and zeros the incidence message.

4. Tier 1: DH²-KT

4.1. Heterogeneous hypergraph construction

Path-derived concept-group hyperedges are bounded-length simple directed paths on audited, acyclic E^{f}_{pre} (Algorithm 1). Each hyperedge collects the concepts visited along a directed path, but Algorithm 1 emits an *unordered* member set, so path direction is not retained in the hyperedge representation. This grouping is used for audit and for optional `HypergraphConv` diagnostic arms; it is not part of the QKC-T / Zero+Δt evaluation arms.

Input: E^{f}_{pre}; bounds $[\ell_{\min}, \ell_{\max}] = [3, 8]$
Output: path-derived concept-group set $\mathcal{E}^{\text{path}}_{f}$
Build directed graph G from E^{f}_{pre}

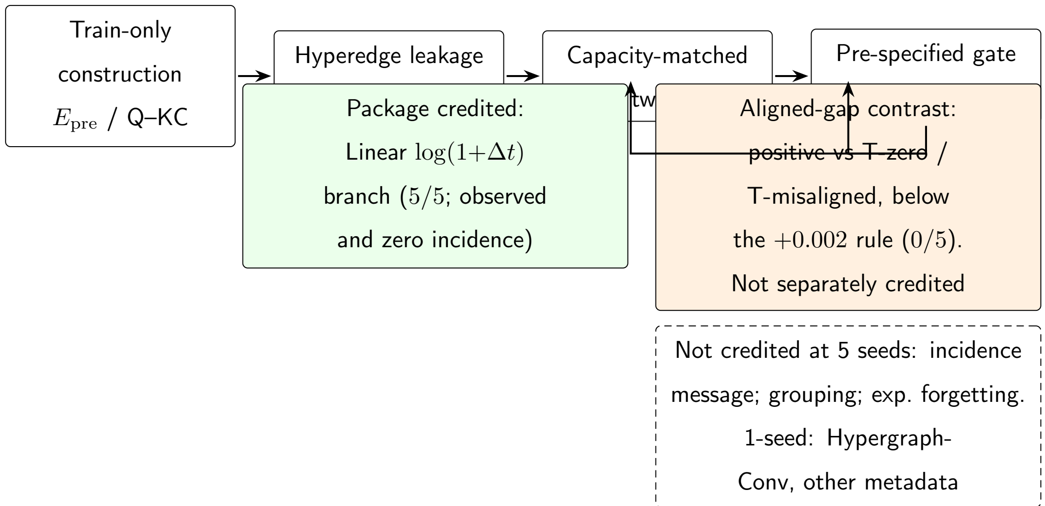

Figure 1: DH^{2}A-KT is an attribution pipeline, not a single predictor. Train-only knowledge is constructed and audited, then compared against capacity-matched twins. The Linear $\log(1{+}\Delta t)$ *package* meets the five-seed gate on XES3G5M; architecture-matched aligned-gap controls stay positive but below the $+0.002$ rule and are not separately credited. The question–KC incidence message does not meet the gate. Teacher grouping and per-concept exponential forgetting are $0/5$. Remaining HypergraphConv/metadata forms in the dashed box are single-seed diagnostics. QKC-T is the observed-incidence evaluation arm; Zero$+\Delta t$ is the credited-minimal positive control. The dashed graph-only encoder and the exploratory Critic survey are not part of this credited map (Appendix A, 6.4).

```
for each root r with in-degree 0 in G do
    for each simple path p from r with ℓ_min ≤ |p| ≤ ℓ_max do
        emit unordered members {(concept, c) : c ∈ p}
    end for
end for
```

Algorithm 1. Path-derived concept-group hyperedges from audited E^{f}_{pre} (member set is unordered; path direction is not encoded). Chain length bounds $[3, 8]$ match `hyperedge.concept_prerequisite` in `configs/xes3g5m.yaml`.

Discussion-thread and teacher-intervention hyperedges are data-blocked on the public KT corpora we use. Session+Hint hyperedges are built on FoundationalASSIST only (Section 6.7). Chain hyperedges on XES3G5M are audited and used in the graph-only diagnostic (Appendix A), not as the QKC-T message-passing graph.

4.2. Hyperedge-level leakage audit

The four pairwise diagnostics of Section 3.1 are reused on the pairwise projection of a hyperedge set. Pairwise TBMR/$|\rho|$ are skipped when that projection exceeds the implementation cap (Table 3). Table 2 adds three classes that a pairwise projection does not express.

Measured values on XES3G5M fold 0 appear in Table 3. To verify that the audit can register leakage when it is present, we add a **full-log positive control**: chain hyperedges built from E_{pre} pooled before the learner split, with a concept→interaction map from the entire log. Under this deliberately contaminated construction, group-membership leak rate reaches 1.0 while the train-only arm stays at 0. Pairwise TBMR on a 10^5-pair subsample is similar across arms (≈ 0.73) and sampled $|\rho|$ stays 0 on both columns, so this positive control validates the *membership* diagnostic, not pairwise $|\rho|$. The zero train-only membership flag therefore reflects protocol design rather than a blind audit.

Table 2: Hyperedge-level leakage taxonomy (extends [8] Table 1)

Class	Source	Diagnostic
Group-membership	A hyperedge with ≥ 1 member drawn from a held-out interaction while other members are train-only	Group-membership leak rate (XES3G5M fold 0: 0; Table 3)
Cross-modal	A Forum/Video embedding pooled over content timestamped after the prediction cutoff	Not instantiated (no Forum/Video logs on the corpora we train)
Intervention-timing	A Teacher/Hint hyperedge whose content was effectively chosen using knowledge of the student's later correctness	Not instantiated for Teacher; Hint members on FoundationalASSIST use train-only session construction (group-membership leak 0)

4.3. Evaluation arms: LSTM with Q←KC and time

The evaluation arm QKC-T is an LSTM backbone with train-only question–KC incidence and a linear $\log(1{+}\Delta t)$ branch; no convolution is applied on the concept stack. We keep QKC-T for GIKT-family parity even though the incidence *message* fails the capacity-matched gate; the credited-minimal control is Zero+Δt (Table 9). (Implementation flags `-question-graph` and `-no-graph` turn incidence on and concept-stack `HypergraphConv` off; Table 4.) Path-derived concept-group chain hyperedges are not consumed on

Table 3: Path-derived concept-group hyperedge audit on XES3G5M fold 0. **Train-only** uses P0's audited E^{f}_{pre} and a train-only concept→interaction map. **Full-log** is a deliberate positive control: E_{pre} pooled before the learner split with a full-log member map (positive control). Pairwise TBMR/$|\rho|$ use a random sample of 10^5 projected pairs (seed 42) when the full projection exceeds 10^5 pairs.

Metric	Train-only	Full-log (pos. control)
Hyperedges	446,305	588,369
ECR^{flag}	0.0	0.0
Group-membership leak rate	0.0	**1.0**
TBMR (sampled)	0.734	0.736
$\lvert\rho\rvert$ (sampled)	0.0	0.0

this path. Let $A \in \mathbb{R}^{N_q \times N_c}$ be the train-only question–KC incidence, row-normalized to $\bar{A}$. Each item q has an embedding e_q; each KC c has e_c. One propagation step, in the spirit of GIKT [3] rather than hypergraph convolution, is

$$\tilde{e}_q = \tanh\big(W(e_q + \bar{A}_{q,:}E_c)\big), \tag{1}$$

the implementation we run (one shared W on the sum, then a separate LSTM input projection). We do not claim a non-collapsible dual-map form. Attribution of the incidence *message* uses a capacity-matched twin that keeps the same question pathway and zeros only $\bar{A}$ (`-question-incidence-control zero`); an older hard-off twin that removed `question_embed` and the projection pathway is capacity-mismatched and is reported only as a pathway-off baseline (Table 7). Interaction embeddings of (c_t, y_t) plus Rasch-style item parameters feed an LSTM. The next step is scored by reading a query vector at the *known* identity of item/KC $t+1$ (standard next-step KT: the model is told *what* will be asked, not whether the learner will be correct).

When `-time-gap` is on, a dedicated map $f(x) = Wx+b$ of $x = \log(1+\Delta t)$ (`nn.Linear(1, hidden)`, bias on) can be added on the LSTM input, the next-step query, or both. A raw gap of 0 therefore yields $f(0) = b$, not a proven zero hidden injection. We treat the prediction time as *attempt-onset correctness prediction*: the score is produced when the learner opens item $t+1$, whose identity is already the query, not when the system first recommends that item. We have not proven that the XES3G5M timestamp is open/start time rather than submit/completion time; query-side availability is therefore a deployment assumption, not a schema fact. LSTM-side Δt uses only gaps between already-observed steps (*history-only*). The query-side gap is the wait until the scored item; that item's identity is part of the prediction request, so the gap is attempt context rather than a post-hoc peek at duration after y_{t+1} is observed. We do not feed the scored step's own response into the query. We report a history-only Δt row (`-time-gap-mode lstm`): five-seed mean Δval+0.00193 (2/5 seeds pass; not credited under the 5/5 rule; Table 15). Query-only: 0/5 seeds pass (mean +0.00099); only `both` meets the rule (5/5). A per-concept exponential forgetting form fails versus Linear on all five seeds (P4; mean Δval$-$0.00252; Table 13). HypergraphConv on multi-KC item hyperedges, hint-item hypergraphs, dual-gated graph-only x_t, and the graph-sensitivity auxiliary loss L_{aux} belong to the v2 diagnostic encoder (Appendix A) or to diagnostic arms (Table 13), not to Eq. (1).

5. Experimental setup

Table 4 maps development labels that appear in logs to the architecture names used in the text and table captions.

5.1. Datasets

Following [8], XES3G5M [23] is the primary predictive benchmark: 18,066 learners, 7,652 items, 865 KCs and 5,549,635 recorded attempts, which expand to 6,413,353 KC-level rows under the standard multi-KC expansion (Section 5.3). ASSISTments 2012 GKT AUC from P0 is not reused as a comparison column. FoundationalASSIST [9] is a secondary fold with synchronized hints (Section 6.7), not interchangeable with classic ASSISTments 2012. Junyi supplies an expert DAG for structural overlap only (Section 6.6); we do not train the full tracer there.

5.2. Baselines reused from P0

KT reference baselines (DKT [1], AKT [24], *simpleKT* [25], GIKT [3], DGEKT [26]) are trained with the vendored pyKT implementations [27] under our clean chunked protocol (Section 5.3). They are a *contextual* benchmark on the same protocol, not a claim that every baseline was tuning-matched to DH^2. We reuse [8]'s published results for BKT [28], GKT [2], and SKT [29] under train-only graph construction (vendored CSVs at commit `5e3d6a0`), except GIKT, AKT, simpleKT, DKT, and DGEKT on XES3G5M, which we retrain under the clean L=400 protocol (Table 7). DyGKT [7] is cited as a related dynamic-graph KT system; it was *not* completed under this evaluator and is not a Table 7 row. GIKT shares DH^2-KT's *training envelope* (batch 16, epoch cap 30, patience 5)—matched limits, not measured

FLOPs or wall-clock. AKT and simpleKT inherit pyKT/P0 defaults with only L raised to 400 (Appendix B; cf. [30]). The companion PDF is available to reviewers as supplementary material on request.

Relation to P0.. Train-only graph construction, edge provenance, and the pairwise DAG audit in [8] are reused and are not contradicted by this paper. Downstream AUC columns in P0 that score every KC-expanded row under an older evaluator are *not* comparable to Table 7 and must not be read as a ranking against QKC-T/Zero+Δt. The ≈8-point graph-only vs. GKT gap in Appendix A mixes encoder, budget, and protocol; it is not an isolated architecture cost. If a P0 claim depends on the older evaluator, that claim should be updated on the P0 manuscript's own track; this paper does not wait on P0 acceptance.

5.3. Clean evaluation protocol

Headline AUCs use a *repeat-target-masked KC-expanded* protocol aligned with pyKT's repeat-format warning [27, 25]: the library's KC-level export expands each multi-KC question into consecutive rows that share the same timestamp and answer, so a large share of eval positions are answerable by copying the input label. We mask repeat-row targets (`-mask-repeats`) in both our models and retrained pyKT baselines, use chunked windows (`-window-mode chunked`) with L=400, and report test-only AUC on the remaining scored KC-rows. This is *not* claimed to be pyKT's question-level all-in-one evaluator [27]: we have not fused KC scores of the same attempt into one p_a. A fold-0 export trace finds 208 chunk windows that start on a globally repeated KC-row (`a2_infoflow_trace.json`); we do not claim that a chunk boundary never promotes a repeat-row into a first target. Table 7 reports DH^2 arms on native clean positions (n=1,093,755) and pyKT baselines on the library intersection (n=1,093,720). Capacity-matched twins (Tables 8–9) and calibration (Table 16) use the native count. An occurrence join on (user, dense-KC) matches 1,093,718 rows; 36 DH^2 rows and 2 pyKT rows remain unmatched, and 8 joined rows disagree on the label (`b6_id_join_summary.json`). The join is not a native attempt ID. Headline five-seed gaps stay Zero+Δt (native) minus the pyKT row (intersection) and are not rebased onto the 1,093,718 join. Learner-level bootstrap CIs in Section 6.2 resample learners on one canonical checkpoint pair; they measure sampling of learners, not training-seed variance, and they do not use the occurrence join (Appendix C). We do not collapse multi-KC rows back to attempt level for the headline AUC: each masked KC-row target is scored once. Table 5 records the already-measured residual between scoring every exported KC-row and scoring the masked remainder (+0.0101 on QKC-T fold 0; +0.0094/+0.0091 on folds 1–2). That residual is the pyKT repeat-format effect [27], not an attempt-level collapse. A no-Δt Q←KC residual of −0.0007 used `window-mode last` and a different L budget (Table 5): it does *not* isolate the time branch from the window rule. An older chunked no-time checkpoint (`v4q_clean`) has residual ≈ -0.0003 and is not the Table 7 Q←KC run. On the fold-0 test export every repeat row has raw $\log(1+\Delta t)=0$ (183,741 rows; 14.4% of unchunked P0-protocol targets, not of the n=1,093,755 scored set). Because $f(0) = b$, that is a zero *feature*, not a proven zero hidden state. We have not rebuilt an attempt-collapsed export or an all-in-one question-level evaluator. The 6,413,353 KC-expanded rows versus 5,549,635 recorded attempts in [23] is the same expansion pyKT warns about. Published P0 numbers that score every KC row are not comparable to Table 7. The retrain uses the shared masking rules; it is not a claim that we discovered KC-level row repeats.

5.4. Credit gate

A knowledge type is *credited* only if every one of five paired seeds has Δval $\geq +0.002$ vs. its matched twin. This is an operating rule locked for this ladder, not a timestamped public registry, and it is *not* the hypothesis $H_0\colon \mu_\Delta = 0$. A paired t-test on Δ answers whether the mean differs from zero; it does not prove $\mu_\Delta > +0.002$. Comparing +0.002 to training-seed SD after the runs is a post-hoc signal-to-noise description, not evidence that the threshold was registered in advance. For Q←KC incidence the matched twin is capacity-matched zero incidence (Table 8): mean Δval$=-0.00010\pm0.00034$ over five seeds (0/5 seeds pass), so incidence content is not credited. A historical hard-off comparison that removed the whole question pathway is capacity-mismatched and is not a credit test. Adding Linear Δt vs. Q←KC has mean +0.00274 (t_4=23.99, two-sided p=1.79×10^{-5}), and adding it on the zero-incidence backbone (Table 9) has mean $+0.00255\pm0.00015$ (5/5 seeds pass). Those twins credit the timing *package*. Same-map T-zero / T-misaligned contrasts remain positive but below the +0.002 rule (Table 10; 0/5). Holm–Bonferroni at α=0.05 is applied to an m=3 family of two-sided paired t-tests on validation Δ (unrounded records in `b8_holm_credit_tests.json`): (i) Linear Δt on zero incidence (t_4=38.01, two-sided $p=2.86\times10^{-6}$, $p_{\text{Holm}}=8.59\times10^{-6}$); (ii) Linear Δt on observed incidence (t_4=23.99, two-sided $p=1.79\times10^{-5}$, $p_{\text{Holm}}=3.58\times10^{-5}$); (iii) incidence observed versus zero (t_4=-0.68, two-sided p=0.53, p_{Holm}=0.53). (i) and (ii) remain significant versus H_0: $\mu_\Delta = 0$; (iii) does not. A previous draft printed Holm values computed on one-sided p beside two-sided raw p; this draft uses one alternative throughout. History-only Δt, P4, and M4 are judged only by the +0.002 5/5 rule, not by this p-family. The family was chosen after the incidence and timing twins existed; we do not present it as frozen before any exploratory run. Table 13 single-seed rows (K, L, M1, M5) are not five-seed credit tests; M5 is a 5k-user Junyi subsample with unordered path-derived groups, not a directed-DAG test. The gate is a credit threshold anchored to seed SD, not a claim that every printed contrast was tested simultaneously.

5.5. Manipulation check

A hypergraph ablation is interpretable only if destroying the constructed hyperedges moves AUC. Near-total node-drop p=0.90 on the graph-only diagnostic encoder (Appendix A) confirms *dependence* on some graph input. Degree-preserving rewiring asks whether that encoder also uses *which* concepts co-occur: it passes on folds 0–2 (ΔAUC 0.0318/0.0341/0.0351; shared clean 0.7485/0.7511/0.7521; Table A.24). Relation-label permutation is near-null on every fold (expected for a single-kind encoder). An earlier pairwise+exercise path was graph-inert (ΔAUC$\approx$ 0.0003). QKC-T and Zero+Δt do not take this check: concept-stack `HypergraphConv` is off, and Q$\leftarrow$KC incidence is a bipartite membership matrix rather than a new multi-way graph.

Table 4: Architecture names used in this paper. Flags in parentheses are implementation aliases from training logs, not model names.

Name in this paper	Architecture
DH^2A-KT	Attribution pipeline: train-only construction, leakage audit, matched twins, credit gate, and knowledge map; not one predictor
QKC-T reference arm	LSTM with train-only Q←KC incidence and Linear $\log(1{+}\Delta t)$; concept-stack HypergraphConv off (`v4`, `-question-graph`, `-no-graph`)
Q←KC only	Same LSTM without the Δt branch
Pathway-off twin	Same LSTM with the question pathway removed (historical hard-off; capacity-mismatched)
Capacity-matched zero twin	Same question pathway with incidence matrix zeroed (`-question-incidence-control zero`)
Zero-incidence+Δt (minimal positive control)	Capacity-matched zero twin plus Linear $\log(1{+}\Delta t)$ at LSTM and query (`zero_time`; T-real)
T-zero / T-misaligned	Same Zero+Δt maps with `-time-gap-control zero` or `misaligned`
Graph-only diagnostic	HypergraphConv encoder on chain hyperedges, no question LSTM path (`v2`)

Table 5: Repeat-mask residual already measured on stored checkpoints (no new training). "P0-protocol" scores every exported KC-row; "clean" masks repeat-row targets (`-mask-repeats`). The +0.0101 residual on QKC-T fold 0 is the drop from scoring all expanded rows to scoring the masked remainder—the pyKT repeat-format effect [27]—not an attempt-level collapse and not a history-side de-duplication. The same recipe on folds 1–2 yields +0.0094 and +0.0091. A Q←KC checkpoint without the Δt branch under a different window rule (`last`, L budget of that run) is near null (−0.0007); that row does *not* isolate the time branch from the window rule. An older chunked no-time residual (`v4q_clean`, ≈ -0.0003) is not the Table 7 Q←KC run. On the fold-0 test export every repeat row has raw Δt=0 (183,741 rows; 14.4% of P0-protocol targets). We have not yet run attempt-level collapse; headline AUCs remain repeat-target-masked KC-expanded, not question-level.

Checkpoint	Window	P0-protocol	Clean	Residual
QKC-T fold 0	chunked	0.8397	0.8296	+0.0101
QKC-T fold 1	chunked	0.8392	0.8298	+0.0094
QKC-T fold 2	chunked	0.8382	0.8291	+0.0091
Q←KC, no Δt (v4q_clean)	last	0.8236	0.8242	−0.0007

Table 6: Two-sided paired t-tests on unrounded validation Δ (`b8_holm_credit_tests.json`; m=3 Holm–Bonferroni). The operating rule is 5/5 seeds with $\Delta\text{val} \geq +0.002$; it is not $H_0\colon \mu_\Delta = 0$ and not $H_0\colon \mu_\Delta \leq 0.002$. Holm is applied to the two-sided raw p (a previous draft printed Holm on one-sided p). History-only, M4, and P4 are judged only by the 5/5 rule.

Contrast	Mean Δval	t_4	Two-sided p	p_{Holm}
Linear Δt on zero incidence	+0.00255	38.01	2.86×10^{-6}	8.59×10^{-6}
Linear Δt on observed incidence	+0.00274	23.99	1.79×10^{-5}	3.58×10^{-5}
Incidence observed vs. zero	−0.00010	−0.68	0.53	0.53

6. Results

6.1. Which structural knowledge predicts (RQ1)

Table 7 reports *test-only* AUC on XES3G5M fold 0 under the clean protocol (mask-repeats, chunked windows, L=400). Table 7 lists fold 0 *test-only* estimates; Table 12 adds learner-fold test-only stability. On five training seeds, the credited-minimal control Zero+Δt reaches 0.8295±0.0001 (n=1,093,755) versus training-budget-matched—not tuning-matched—GIKT 0.8258±0.0001, AKT 0.8223±0.0001, and *simpleKT* 0.8214±0.0007 (n=1,093,720; unrounded mean gaps +0.0036, +0.0072, and +0.0081). Q←KC without time is 0.8268±0.0002 on the same native test-only split. QKC-T is 0.8295±0.0001 over the same five seeds (seed 42 0.8296 matches the calibration checkpoint); the archived A2 combined val+test mean 0.8296±0.0001 (n=1,640,242) is not used as a Table 7 cell. Retrained DKT and DGEKT are 0.8009±0.0002 and 0.7850±0.0001. Under the capacity-matched zero-incidence twin (Table 8), the incidence *message* does not meet the gate on any seed (0/5; mean Δval=$-$0.00010±0.00034). Adding Linear log(1+Δt) on the observed-incidence backbone meets the val rule 5/5 (Table 15); we credit that within-backbone package, not a combined-AUC gap versus GIKT. We do not credit hypergraph convolution for these numbers: that arm is `-no-graph` on the concept stack. The same QKC-T recipe on folds 1 and 2 stays within 0.001 of fold 0 (mean 0.8295±0.0004 over learner folds, test-only; Table 12).

Table 13 lists pre-specified diagnostics (Δval gate +0.002, not loosened). HypergraphConv on multi-KC item hyperedges (K), hint-item session hyperedges on FoundationalASSIST (L), a full Q-matrix (M1), and a Junyi expert

DAG on a 5k-user subsample (M5) remain single-seed. Teacher grouping (M4; ASSIST2012 L=200) and per-concept exponential forgetting (P4; XES L=400) are now five-seed and do not meet the rule on any seed (0/5; mean Δval +0.00025 and −0.00252). "Does not meet the gate" means the form is not credited in that configuration, not that the knowledge type is valueless. The small FoundationalASSIST session-HE shift 0.720 → 0.722 (concept projection; Table 20) is therefore not a hypergraph-AUC claim: the isolated hint-item HE arm did not pass.

6.2. Time as attempt context (RQ2)

On XES3G5M, adding Linear Δt into both LSTM and next-step query meets the 5/5 operating rule against Q←KC (Δval $\geq$ +0.002; two-sided Holm p=3.58×10^{-5} in the m=3 family of Section 5.4). This is a time-*branch* contrast. Architecture-matched T-zero / T-misaligned controls that keep the same maps and change only the gap input do not meet the val rule (0/5; mean Δval+0.00183 / +0.00144; mean Δtest-only+0.0020 / +0.0015; Table 10). Adding the time branch improves the ablations. We credit that package under the pre-specified rule; we do not isolate aligned gap input as a separately credited source. Five-seed test-only QKC-T versus Q←KC is 0.8295±0.0001 vs. 0.8268±0.0002 on native n=1,093,755 (Table 7); seed 42 calibration is 0.8296 vs. 0.8270, and the learner-level 95% CI on that canonical pair is $[+0.00225, +0.00298]$. The same timing branch also passes when the backbone has *zero* Q–KC incidence (5/5 seeds pass; mean Δval+0.00255, mean Δtest-only+0.00269; Table 9), so the timing credit is not an artefact of nonzero incidence. A quartile slice of log(1+gap) collapses to two bins: short gaps ($\lesssim$18 min) Δ+0.0028, long gaps Δ+0.0019, both CIs excluding 0

— not a long-horizon exponential-decay pattern under P4's form (0/5 seeds pass vs. Linear). Table 15 ablates injection site on XES3G5M. History-only LSTM Δt (observed gaps, no query-side wait) reaches a five-seed test-only AUC 0.8286±0.0002 (Table 15) and is not credited (mean Δval+0.00193, 2/5 seeds pass); query-only is 0/5 (mean +0.00099; test-only 0.8278±0.0002); only `-time-gap-mode both` meets the rule on all five seeds (mean +0.00274). The branches are complementary: QKC-T's timing credit is not available under a history-only deployment. On FoundationalASSIST, query-only Δt keeps most of the M2 gain (+0.01471 vs. off) while LSTM-only is +0.00280; combining time with prior-step `saw_answer` is additive (+0.00297 vs. M2).

Per-concept $\exp(-\mathrm{softplus}(\theta_c) \cdot \mathrm{gap})$ does not meet the gate versus Linear on any seed (P4; mean Δval$-$0.00252). We treat Linear Δt as attempt-context knowledge on this backbone, not as a hypergraph effect, and we do not generalise that to every forgetting form. Query-side Δt uses the gap to the item whose identity is already the next-step query; it is not the duration of a step whose correctness has been observed. Against AKT at the same L=400 clean protocol (test-only 0.8223±0.0001 over five seeds), the five-seed test-only mean gaps from Zero+Δt are +0.0072 versus AKT and +0.0081 versus *simpleKT* (unrounded means; printed cells subtract to the same hundredths). As a secondary diagnostic, a paired learner-level bootstrap (1000 resamples, 3614 learners) on one canonical fold-0 prediction file—not the subtraction of the five-seed ± rows—gives 95% CIs $[+0.0035, +0.0043]$ vs. GIKT, $[+0.0068, +0.0077]$ vs. AKT, and $[+0.0088, +0.0098]$ vs. *simpleKT* (Table C.26). Those intervals measure learner sampling on a fixed checkpoint pair; they do not quantify training-seed variance (Table 7).

6.3. Calibration

Table 16 reports 15-bin ECE, Brier score, and NLL on fold 0 test predictions under the same clean protocol. Among DH^2 arms QKC-T is best calibrated (ECE 0.0056) despite higher AUC; Zero+Δt is 0.0072 on the same native n; Q←KC on is slightly miscalibrated high (ECE 0.0103). On the pyKT intersection, AKT has the lowest ECE (0.0047); GIKT is 0.0073 and *simpleKT* 0.0100. No reported ECE exceeds 0.05. We report calibration as a diagnostic, not as a primary claim.

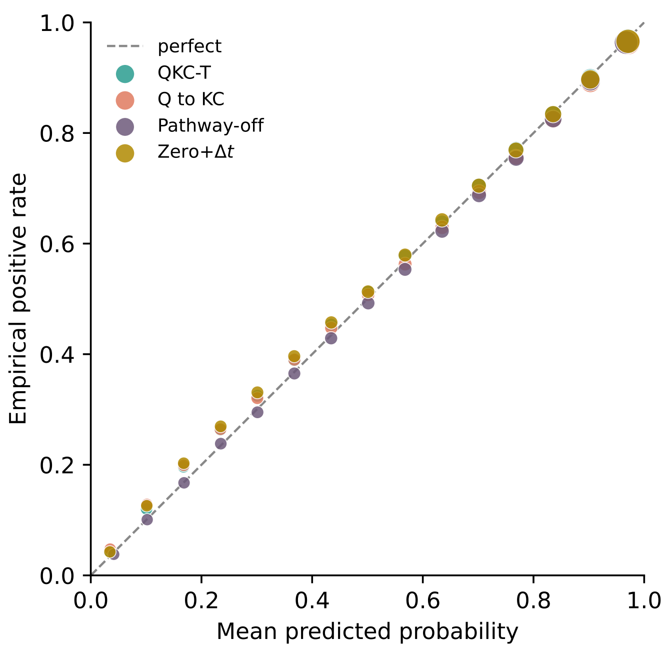

Figure 2: Fifteen-bin reliability on XES3G5M fold 0 under the clean protocol (mask-repeats, chunked, L=400; seed 42 checkpoints). Marker area scales with bin count. Labels: QKC-T, Q←KC, Pathway-off, Zero+Δt (ECE 0.0056 / 0.0072 on QKC-T / Zero+Δt; Table 16). pyKT baselines were not re-scored here.

6.4. Exploratory pilot: Critic-gated explanations

This subsection is an exploratory pilot after the Tier 1 results, not a contribution. Three agents read frozen DH^2-KT scores: a Diagnostician writes a rationale, a Critic flags output that contradicts the frozen P(correct) (15-bin ECE 0.0056 on QKC-T; Table 16), and a Tutor writes session hyperedges that the frozen tracer does not consume. We ran n=500 prompts (XES3G5M fold 0, seed 42, Qwen2.5-7B via Ollama) on the graph-only v2 checkpoint and again on QKC-T. Following [15], we report Critic flag rate and ID-anchored KC-Jaccard. Independent generation (IG) omits P(correct) from the Diagnostician prompt while the Critic still sees the true score, so the flag-rate gap is largely definitional and is not a content finding. On QKC-T, grounded/IG flags are 0.018/0.766 with mean JC 0.161/0.152; the JC gap is $\approx +0.010$ (Table 17, Table 18). Blinded Likert ratings on a seed 42 subsample of n=100 grounded tracer explanations (3 independent raters) give faithfulness 4.55±0.63 and usefulness 4.01±0.74 (ordinal Krippendorff α=0.52/0.56; mean pairwise Spearman ρ=0.43/0.34; within±1 95.0%/94.7%; Table 19). An earlier dual-rater n=40 pack on the v2 graph-only checkpoint is archived and not the primary human-eval row.

6.5. Paired diagnostic knobs

Table 20 pairs remaining diagnostic knobs with the pilot's faithfulness knobs. Removing the Critic sets the flag rate to 0 by construction while leaving ID-anchored KC-Jaccard unchanged. Dropping L_{aux} (`graph_sensitivity_weight` = 0) on the graph-only fold 0 track still yields clean AUC 0.754 and ΔAUC= 0.039 (pass); that track is the appendix diagnostic, not Table 7.

6.6. Ground-truth structural overlap (Junyi)

Table 21 compares hyperedge-chain overlap with P0's pairwise E_{pre} against the Junyi expert DAG at $K{=}|\mathrm{expert}|$. Edge F1 is 0.165 (hyperedge chain) vs. 0.183 (E_{pre}) on fold 0; folds 1–2 are analogous. The chain representation slightly *underlaps* pairwise E_{pre}: collapsing directed paths into multi-node hyperedges, then projecting back to pairs for overlap scoring, can dilute expert-edge recall relative to scoring E_{pre} directly. We treat this as structural validation of construction fidelity, *not* as evidence that chain hyperedges improve predictive KT on Junyi — full DH^2-KT training on Junyi (25M interactions) was not run.

6.7. Secondary corpus: FoundationalASSIST

FoundationalASSIST [9] provides ≈1.74M skill-linked interactions with synchronized `hint_count` (hint-use rate 5.2%). Using the same learner-temporal protocol as P0 (0.7/0.1/0.2, fold 0), we infer train-only E_{pre} (280 edges after cycle pruning), build 5,840 path-derived concept-group chain hyperedges, and construct 308,670 session hyperedges on train users (11.6% with a Hint member; $\mathrm{ECR}^{\mathrm{flag}}{=}0$, group-membership leak=0). DH^2-KT fold 0 AUC is 0.720 with path-derived concept-group hyperedges alone ($n{=}298{,}500$ predictions). Adding up to 20k multi-concept session hyperedges (Hint-preferring selection; concept co-occurrence projection) raises AUC to 0.722 ($\Delta{+}0.002$) — a small positive shift under the same local budget, not a matched GKT comparison. Table 22 summarizes the corpus and those AUCs. An IPW identification check of hint use, with weak positivity, is in Appendix D; it is not a contribution.

Table 7: Test-only AUC on XES3G5M fold 0 under the *clean* protocol (mask-repeats, chunked windows, L=400). DH^2 rows use native scoring (n=1,093,755). pyKT baselines use the library intersection (n=1,093,720; occurrence join in Section 5.3). Values are mean ± sample SD over five training seeds (42, 17, 1234, 0, 2024) of the *same* test-only metric. QKC-T seed 42 is 0.829593 and matches Table 16. The archived A2 combined val+test mean 0.8296±0.0001 (n=1,640,242) is *not* this column. Zero+Δt is the credited-minimal control (Table 9); QKC-T is the observed-incidence evaluation arm for GIKT-family parity. Unrounded five-seed means agree to five decimals (QKC-T 0.829492; Zero+Δt 0.829486). GIKT is matched to the DH^2 *training envelope* (batch 16, cap 30 epochs, patience 5), not to a hyperparameter search (Appendix B). Five-seed mean gaps from unrounded test-only means: QKC-T (or Zero+Δt) minus GIKT +0.0036, minus AKT +0.0072, minus *simpleKT* +0.0081 (printed cells subtract to the same hundredths). History-only Δt and the capacity-mismatched pathway-off twin are not in this table (Table 15; Section 5.4). This is *not* the three-fold graph-only v2 table (Appendix A).

Model	Test-only AUC	n scored	Seeds
DH^2-KT QKC-T (eval. arm)	0.8295±0.0001	1,093,755	5
DH^2-KT Zero+Δt (credited)	0.8295±0.0001	1,093,755	5
DH^2-KT Q←KC	0.8268±0.0002	1,093,755	5
GIKT (training-budget-matched)	0.8258±0.0001	1,093,720	5
AKT (L=400)	0.8223±0.0001	1,093,720	5
simpleKT (L=400)	0.8214±0.0007	1,093,720	5
DKT (L=400)	0.8009±0.0002	1,093,720	5
DGEKT (L=400)	0.7850±0.0001	1,093,720	5

Table 8: Capacity-matched Q←KC attribution on XES3G5M fold 0 (clean protocol). Both arms keep `-question-graph` (same `question_embed`, GCN linear, and input projection); only the train-only incidence matrix is replaced by zeros in the twin (`-question-incidence-control zero`). Pre-specified gate: Δval $\geq +0.002$. Checkpoint state shapes match on every seed; observed nonzero links = 8848, zero arm = 0. Mean Δval=-0.00010 ± 0.00034; mean Δtest-only$\approx+0.00004$. 0/5 seeds pass: the incidence *message* is not credited. The older hard-off twin that removed the whole question pathway is not used for this claim (Section 5.4). Δtest columns are outer test-only AUC (1,093,755 scored positions).

Seed	Observed val	Zero val	Δval	Δtest-only	Gate
0	0.82554	0.82610	−0.00056	−0.00007	no
17	0.82579	0.82589	−0.00010	−0.00018	no
42	0.82598	0.82619	−0.00021	+0.00010	no
1234	0.82605	0.82567	+0.00038	+0.00014	no
2024	0.82608	0.82610	−0.00003	+0.00020	no

Table 9: Linear Δt on the capacity-matched *zero-incidence* backbone (XES3G5M fold 0, clean protocol). Both arms keep zero Q–KC incidence; `zero_time` adds LSTM+query $\log(1+\Delta t)$ (`-time-gap-mode both`). Pre-specified gate: Δval $\geq +0.002$. Mean Δval=$+0.00255\pm0.00015$; mean Δtest-only=$+0.00269\pm0.00012$. 5/5 seeds pass: the *branch* credit is not an artefact of nonzero incidence. State shapes differ by the timing parameters (expected).

Seed	Zero val	Zero+Δt val	Δval	Δtest	Gate
0	0.82610	0.82852	+0.00242	+0.00270	yes
17	0.82589	0.82835	+0.00245	+0.00258	yes
42	0.82619	0.82862	+0.00243	+0.00255	yes
1234	0.82567	0.82837	+0.00270	+0.00282	yes
2024	0.82610	0.82882	+0.00272	+0.00280	yes

Table 10: Architecture-matched time-*input* controls on the zero-incidence Linear-Δt backbone (XES3G5M fold 0, five training seeds). All three arms keep the extra Linear maps. T-real reuses the five Zero+Δt checkpoints (not retrained). T-zero trains with all gaps set to 0; T-misaligned permutes non-start gaps within each learner. The operating rule is Δval $\geq +0.002$ vs. T-real on every seed: 0/5 vs. T-zero (mean +0.00183) and 0/5 vs. T-misaligned (mean +0.00144). Test-only means remain positive (+0.0020 / +0.0015). We credit the timing *package* under the pre-specified rule; we do not credit aligned gap input as a separate source.

Seed	Δval 0	Δval mis.	Gate	Δtest 0	Δtest mis.
0	+0.00179	+0.00148	no	+0.00201	+0.00157
17	+0.00187	+0.00124	no	+0.00193	+0.00156
42	+0.00180	+0.00159	no	+0.00170	+0.00140
1234	+0.00175	+0.00108	no	+0.00220	+0.00147
2024	+0.00196	+0.00179	no	+0.00192	+0.00147

Table 11: Leakage-audited knowledge attribution summary. The upper block is five paired training seeds on XES3G5M (or ASSIST2012 for teacher grouping) and is the credit map. The lower block is single-seed diagnostics and is *not* a five-seed credit test. "Observed" and "zero" refer only to the Q–KC incidence message; both retain the same question embedding and projection pathway. The operating rule is Δval $\geq +0.002$ versus the stated twin on every seed (5/5). Timing *branch* rows add maps as well as gap input; the aligned-input row keeps those maps and changes only the gap (Table 10).

Knowledge message	Matched twin	Corpus / seeds	Mean Δval	Credit
Five-seed credit tests				
Q←KC incidence	Zero incidence	XES / 5	$-0.00010{\pm}0.00034$	0/5; not credited
Linear Δt (both)	No Δt, observed incidence	XES / 5	$+0.00274{\pm}0.00025$	5/5; credited package
Linear Δt (history-only)	No Δt, observed incidence	XES / 5	+0.00193	2/5; not credited
Linear Δt (both)	No Δt, zero incidence	XES / 5	$+0.00255{\pm}0.00015$	5/5; credited package
Aligned Δt input	T-zero (same maps)	XES / 5	+0.00183	0/5; below rule
Teacher grouping	Off-group twin	ASSIST2012 / 5	+0.00025	0/5; not credited
Per-concept exponential gap	Linear Δt	XES / 5	$-0.00252{\pm}0.00019$	0/5; not credited
Single-seed diagnostics (not five-seed credit)				
HypergraphConv	No concept convolution	XES / 1	+0.00025	below gate
Hint-item session HE	No hint-item HE	FA / 1	+0.00007	below gate
Full Q-matrix incidence	Matched control	FA / 1	−0.00010	below gate
Expert prerequisite DAG	Matched control	Junyi 5k / 1	+0.00056	below gate

Table 12: QKC-T *test-only* AUC on XES3G5M under the clean protocol (mask-repeats, chunked, L=400). Rows 0–2 are one checkpoint per learner fold (P0 split seeds); mean±SD is over folds, n is native DH^2 test positions. Fold 0 here is the seed-42 checkpoint (0.8296); the five-seed test-only mean 0.8295±0.0001 is Table 7. The archived A2 combined val+test mean 0.8296±0.0001 (n=1,640,242) is *not* this column.

Fold	Test-only AUC	n scored
0	0.8296	1,093,755
1	0.8298	1,092,664
2	0.8291	1,090,374
Mean±SD (folds)	**0.8295±0.0004**	—

Table 13: Negative results under a pre-specified validation gate (Δval $\geq$ +0.002 vs. a matched twin). "Below gate" means the form is not credited in the configuration we ran, not that the knowledge type is valueless. All runs keep a separate Linear/embedding branch (no absorption into `concept_embed`). Hypergraph convolution and hint-item hyperedges are among these diagnostics; they are not AUC novelty. M4 and P4 are five-seed means (seeds 42, 17, 1234, 0, 2024; 0/5 seeds pass); P4's twin is same-seed Linear Δt / QKC-T (Table 14), not Zero+Δt. K, L, M1, and M5 remain single-seed and are not five-seed credit tests. M5 is a 5,000-user Junyi subsample with path-derived *unordered* concept groups (Algorithm 1), not a directed-DAG test on the full corpus.

ID	Knowledge (corpus)	Twin val	ON val	Δval	Gate
K	HypergraphConv, multi-KC HE (XES)	0.82598	0.82623	+0.00025	below gate
L	Hint-item session HE (FA)	0.80861	0.80868	+0.00007	below gate
M1	Full Q-matrix incidence (FA)	0.80834	0.80824	−0.00010	below gate
M4	Teacher grouping (ASSIST2012, 5-seed)	0.76475	0.76499	+0.00025	0/5; not credited
M5	Expert prereq. DAG (Junyi 5k)	0.7640	0.76456	+0.00056	below gate
P4	Per-concept exp. forgetting (XES, 5-seed)	0.82862	0.82610	−0.00252	0/5; not credited

Table 14: Per-concept exponential forgetting (P4) versus Linear Δt on the QKC-T backbone (XES3G5M fold 0, five paired seeds). Twin is same-seed `-time-gap-mode both` (QKC-T), not Zero+Δt. Mean Δval=-0.00252 ± 0.00019 (0/5 seeds pass). The printed five-seed mean twin 0.82862 coincides at five decimals with Zero+Δt seed 42 val in Table 9; both values come from different files and are not a copied cell.

Seed	Twin val (QKC-T)	P4 val	Δval	Gate
0	0.82860	0.82596	−0.00263	no
17	0.82858	0.82587	−0.00270	no
42	0.82878	0.82619	−0.00258	no
1234	0.82841	0.82618	−0.00223	no
2024	0.82875	0.82630	−0.00245	no
mean	0.82862	0.82610	−0.00252	0/5

Table 15: Query-side vs. history-only Linear Δt on XES3G5M fold 0 (clean L=400, five training seeds). Twin is Q←KC without time-gap. Pre-specified gate: Δval $\geq +0.002$ (validation only; this column is the credit test). The AUC column is the same *test-only* metric as Table 7 (n=1,093,755). `both` is QKC-T (five seeds). Only `both` meets the val gate on all five seeds. History-only is 2/5 seeds pass and is not credited under the 5/5 rule.

Mode	Injection site	Test-only AUC	Gate (k/5)
`both`	LSTM + query	0.8295±0.0001	5/5 (mean Δval +0.00274)
`lstm` (history-only)	LSTM only (observed gaps)	0.8286±0.0002	2/5 (mean Δval +0.00193)
`query`	next-step query only	0.8278±0.0002	0/5 (mean Δval +0.00099)

Table 16: Calibration on XES3G5M fold 0 test predictions under the clean protocol (mask-repeats, chunked, L=400). ECE uses 15 equal-width bins; lower is better. DH^2 checkpoints: seed 42 (QKC-T AUC 0.8296 here matches Table 7; Zero+Δt 0.8294 is the same seed-42 checkpoint as Table 9; the pathway-off twin 0.8202 is calibration-only and is not a Table 7 row), scored on native n=1,093,755 positions. pyKT GIKT/AKT/*simpleKT* use the same seed-42 prediction files as the headline intersection (n=1,093,720). An occurrence join on (user, dense-KC) matches 1,093,718 rows; 36 DH^2 rows and 2 pyKT rows remain unmatched (`b6_id_join_summary.json`). We report calibration as a diagnostic and do not apply temperature scaling. Reliability curves (DH^2 arms, including Zero+Δt) appear in Figure 2.

Model	AUC	Brier	NLL	ECE
Q←KC +Δt	0.8296	0.1207	0.3828	0.0056
Zero+Δt	0.8294	0.1208	0.3831	0.0072
Q←KC on	0.8270	0.1216	0.3857	0.0103
Q←KC off (twin)	0.8202	0.1235	0.3911	0.0081
GIKT (training-budget-matched)	0.8257	0.1219	0.3864	0.0073
AKT (L=400)	0.8223	0.1229	0.3893	0.0047
simpleKT (L=400)	0.8203	0.1236	0.3916	0.0100

Table 17: Exploratory Critic flag rates (XES3G5M fold 0, n=500, seed 42, Qwen2.5-7B via Ollama unless noted). Early rows are historical graph-only packs; the QKC-T grounded/IG pair is the reported-checkpoint rerun (Table 18). This table is not a predictive attribution result.

Run	Backend	Flag rate	Mean P(correct)
Stub (sanity)	StubLLM	0.000 (0/500)	0.742
Ollama graph-only (first pack)	qwen2.5:7b	0.124 (62/500)	0.746
Ollama graph-only (format calib.)	qwen2.5:7b	0.000 (0/500)	0.753
Grounded + `GroundedKCs` (graph-only)	qwen2.5:7b	0.026 (13/500)	—
IG (graph-only, w/o P in Diag.)	qwen2.5:7b	0.768 (384/500)	—
Grounded + Critic (QKC-T)	qwen2.5:7b	0.018 (9/500)	—
IG (QKC-T)	qwen2.5:7b	0.766 (383/500)	—

Table 18: Tier 2 explanation faithfulness: grounded Diagnostician vs. independent generation (IG). ID-anchored KC-Jaccard (JC) measures overlap between `GroundedKCs` trailer IDs and the support set $\{\text{target}\} \cup \text{history} \cup \text{1-hop } E_{\text{pre}}$. IG omits frozen P(correct) from the Diagnostician prompt; the Critic still sees the true Tier-1 score. The original Ollama pack used the graph-only diagnostic checkpoint. The reported QKC-T checkpoint is a same-prompt re-run (n=500).

Arm	Backbone	Backend	n	Flag	Mean JC
Grounded	—	StubLLM	20	0.000	0.191
IG	—	StubLLM	20	1.000	0.191
Grounded	graph-only diagnostic	qwen2.5:7b	500	0.026	0.152
IG	graph-only diagnostic	qwen2.5:7b	500	0.768	0.142
Grounded	QKC-T	qwen2.5:7b	500	0.018	0.161
IG	QKC-T	qwen2.5:7b	500	0.766	0.152

Table 19: Human rating of the exploratory Tier 2 pilot from the QKC-T checkpoint (XES3G5M fold 0, grounded Ollama subsample, n=100 explanations, 3 independent raters, 1–5 scales). Agreement: ordinal Krippendorff α and the mean of pairwise Spearman ρ / fraction of scores within ± 1.

Scale	Mean $\pm$ SD	Agreement (3 raters)
Faithfulness	4.55 ± 0.63	α=0.52; ρ=0.43; within$\pm$1: 95.0%
Usefulness	4.01 ± 0.74	α=0.56; ρ=0.34; within$\pm$1: 94.7%

Table 20: Paired ablation of DH^{2}A-KT knowledge and explanation knobs. Predictive rows use the clean L=400 protocol (fold 0) except HypergraphConv / FA session. QKC-T, Q$\leftarrow$KC, and Zero+Δt $\pm$ cells are five-seed test-only (Table 7). The 0.8296 Critic row is the seed-42 checkpoint used for the n=500 prompt pack. The pre-specified gate is Δval $\geq$ +0.002. Manipulation ΔAUC is the graph-only diagnostic (Appendix A), not QKC-T.

Setting	AUC	Gate / note	Flag	Mean JC
QKC-T (eval. arm, 5-seed)	0.8295±0.0001	val gate 5/5 +0.00274	—	—
Q$\leftarrow$KC (5-seed test-only)	0.8268±0.0002	not an incidence credit	—	—
Zero+Δt (credited, 5-seed)	0.8295±0.0001	val gate 5/5 +0.00255	—	—
HypergraphConv (val)	0.8262	below gate +0.00025	—	—
Grounded + Critic (graph-only diagnostic)	0.752±0.002	manip. fold 0 ΔAUC 0.0323 (B9)	0.026	0.152
IG (graph-only diagnostic)	—	—	0.768	0.142
Grounded + Critic (QKC-T)	0.8296	same prompt, n=500	0.018	0.161
IG (QKC-T)	—	—	0.766	0.152
w/o Critic	—	—	0.000	0.152
FA: concept-only	0.720	—	—	—
FA: + session HE	0.722	L below gate +0.00007	—	—

Table 21: Ground-truth cross-validation on Junyi Academy (fold 0, top-K=|expert|=1131). Precision equals recall (and therefore F1) by construction because K is set to the number of expert edges. Hyperedge chain slightly underlaps pairwise E_{pre} but remains the same order of magnitude.

Representation	Edge F1	Edge prec.	Edge rec.
P0 E_{pre}	0.183	0.183	0.183
Hyperedge chain	0.165	0.165	0.165

Table 22: FoundationalASSIST secondary corpus (fold 0). Session hyperedges use a 30-minute gap; Hint members appear when `hint_count`> 0. Predictive AUC is observational (no P0-reused GKT/simpleKT column). An IPW identification check of hint use is in Appendix D; it is not a contribution.

Quantity	Value
Interactions (canonical)	1,740,987
Learners / items / KCs	5,000 / 3,359 / 209
Hint-use rate	0.052
Train-only E_{pre} edges (after cycle prune)	280
Path-derived concept-group chain hyperedges	5,840
Session hyperedges (train users)	308,670
with $\geq$1 Hint member	35,752 (11.6%)
Session hyperedge ECR$^{\text{flag}}$ / group leak	0 / 0
DH^2-KT fold 0 AUC (path-derived concept-group)	0.720
+ session HE (20k, Hint-prefer)	0.722 (Δ+0.002)
n next-step predictions	298,500

7. Discussion

7.1. Threats to validity

Headline AUCs (Table 7) are clean L=400, test-only. QKC-T and Zero+Δt are both five-seed test-only (0.8295±0.0001); three-fold QKC-T stability is in Table 12. Training-budget-matched GIKT/AKT remain fold 0. Other baselines come from reused P0 tables on the older budget. The GIKT edge is small (+0.0036 from Zero+Δt) and is a matched *training envelope*, not a matched tuning search (Appendix B). Learner-level bootstrap CIs on one checkpoint are a secondary diagnostic of learner sampling, not training-seed CIs, and they do not use the 1,093,718-row occurrence join. HypergraphConv and hint-item hyperedges did not meet the single-seed gate, and teacher grouping / exponential forgetting are 0/5, so we do not credit multi-way message passing from those arms. The node-drop check (Appendix A) is run on the graph-only diagnostic encoder, not on QKC-T. Node-drop confirms *dependence* on some graph input. Degree-preserving rewiring on folds 1–2 additionally moves AUC (0.0341/0.0351), which is evidence that this diagnostic encoder is sensitive to co-occurrence structure, not only to the presence of nodes (shared-clean AUCs in Table A.24). That check still does not support graph-structure claims about QKC-T. Q←KC incidence is a bipartite membership matrix, not a new multi-way graph.

Deployment availability of the query-side gap.. Attempt-onset scoring (this paper's protocol) produces $P(y_{t+1}{=}1)$ when the learner opens item $t{+}1$, so the query-side wait has already elapsed. History-only / recommend-before-return uses only LSTM-side gaps already in the log; that arm is the history-only row (five-seed mean Δval, 2/5 seeds pass; not credited; Table 15). Serving a next item before the learner returns is earlier still: the wait has not occurred, so query-side Δt must be omitted or replaced by a planned-gap proxy we do not evaluate. The credited timing arm therefore assumes attempt-onset availability; a history-only deployment should use the LSTM-only arm and must not claim the both-mode 5/5 credit (mean Δval+0.00274).

7.2. Interpretation of the empirical findings

The results answer an attribution question, not "hypergraph NNs beat GIKT." Each paragraph below asks whether one knowledge type survived the pre-specified ablation, and what that means for a system builder.

Question–KC incidence.. Does not survive as credited structural knowledge under the capacity-matched twin (0/5 seeds pass; Table 8). The QKC-T evaluation arm still includes the GIKT-class question pathway for architecture parity, but the train-only incidence message itself is not what the gate attributes.

Linear $\log(1+\Delta t)$.. Survives on this backbone when both LSTM-side and query-side gaps are present (5/5 vs. Q←KC; 5/5 vs. zero incidence; Table 9). Those contrasts add maps as well as gap input. T-zero / T-misaligned keep the maps: the aligned-gap contrast is positive but below the val rule (Table 10). History-only Δt does not meet the 5/5 rule (Table 15): the credited package uses attempt-onset scoring, not a history-only forgetting cue.

Hypergraph convolution and path-derived concept groups.. Did not meet the single-seed gate: `HypergraphConv` on chain hyperedges is not credited as five-seed evidence (Table 13). Those groups remain an audit/diagnostic artefact.

The ≈8 point gap versus GKT mixes encoder, budget, and protocol; it is not an isolated architecture cost of QKC-T.

Hint-item hyperedges, full Q-matrix, teacher grouping, expert DAG, exponential forgetting.. Hint-item HE, full Q-matrix, and the Junyi DAG did not meet the single-seed gate. Teacher grouping and per-concept exponential forgetting are 0/5 (Table 13). Session+Hint on FoundationalASSIST (0.720 → 0.722) is a small local shift, not credited as multi-way graph novelty.

The Critic pilot in Section 6.4 is exploratory and is not part of this attribution map.

7.3. Limitations

- Discussion-thread and teacher-intervention hyperedges remain data-blocked; session+Hint hyperedges on FoundationalASSIST shift AUC 0.720 → 0.722 under concept projection, but the pre-specified hint-item hypergraph arm does not meet the +0.002 gate (L in Table 13) and is not claimed as hypergraph novelty.

- Appendix D reports an IPW hint check with weak positivity; it is not a contribution and must not be read as a recommendation against hints.

- Teacher grouping (M4) and per-concept exponential forgetting (P4) do not meet the +0.002 gate on any seed (mean Δval +0.00025 and −0.00252). The remaining four single-seed rows in Table 13 (K, L, M1, M5) are diagnostics below the gate, not five-seed credit tests.

- Calibration (ECE/Brier/NLL) is reported for QKC-T, Zero+Δt, Q←KC, and a historical pathway-off twin (native n=1,093,755) and for GIKT/AKT/*simpleKT* on the pyKT intersection (n=1,093,720), all seed 42 (Zero+Δt ECE 0.0072). We report calibration as a diagnostic and do not treat ECE<0.05 as a universal certification. History-only and query-only test-only means now use all five seeds (Table 15). The archived A2 combined val+test mean 0.8296±0.0001 (n=1,640,242) is not a Table 7 cell.

- Repeat-mask residual on QKC-T is +0.0101 on fold 0 (Table 5). On the fold-0 test export, every repeat row has raw Δt=0 and $\log(1+\Delta t)=0$ (183,741 rows; 14.4% of P0-protocol targets; `a3_dt_repeat_dist.json`). Headline AUCs remain mask-target; 208 chunk windows start on a globally repeated KC-row. We have not run attempt-level collapse or a question-level all-in-one evaluator. The DH^2/pyKT occurrence join leaves 36+2 unmatched rows and 8 label mismatches; headline n is not rebased onto the join.

- We do not evaluate a planned-gap proxy for query-side Δt (the wait until a recommended item is actually opened). Deployment that scores before the learner returns must omit query-side Δt (Section 7.1).

- The hyperedge leakage audit's pairwise-projection approximation (Section 4.2) weights all projected pairs equally; the full-log control moves group-membership leak to 1.0 but not sampled $|\rho|$. A hyperedge-native statistic is future work.

- Exploratory Critic pilot: one fold, one local 7B LLM. Blinded Likert on QKC-T (n=100, 3 raters) is reported in Table 19; ordinal Krippendorff α is 0.52 (faithfulness) / 0.56 (usefulness). Mean pairwise Spearman is moderate (ρ≈0.43/0.34) while within±1 agreement is high (≈95%), consistent with compressed high scores. An earlier dual-rater n=40 pack on the graph-only diagnostic checkpoint is archived only. The grounded–IG flag gap is largely definitional because IG withholds P(correct) from the Diagnostician while the Critic scores against that number. ID-anchored KC-Jaccard is not synonym-aware (XES3G5M lacks an in-repo KC name vocabulary); the JC gap is ≈ +0.010. An LLM-scale sensitivity sweep remains optional future work.

- Reproduction environment: Windows workstation, single RTX 3090; unit tests pass in the released codebase.

8. Conclusion and future work

We presented DH^{2}A-KT as leakage-audited knowledge attribution: which structural and temporal knowledge survives a controlled ablation. Adding a Linear Δt *branch* meets the five-seed operating rule on the QKC-T backbone and on a capacity-matched zero-incidence twin; capacity-matched zeroing of the question–KC incidence message does not. Architecture-matched T-zero / T-misaligned controls on the same maps remain positive but below the +0.002 rule (0/5). We credit the timing package, not aligned gap input as a separate source. QKC-T is the observed-incidence evaluation arm; Zero+Δt is the credited-minimal control. On XES3G5M (repeat-target-masked KC-expanded, L=400, test-only), both arms reach 0.8295±0.0001 over five seeds. Table 7 is a contextual benchmark: training-budget-matched GIKT is 0.8258±0.0001. Three learner folds of QKC-T mean 0.8295±0.0004.

History-only Δt is 2/5 seeds pass and is not credited. Future work: a question-level all-in-one evaluator; five-seed repeats of K/L/M1/M5; and hyperedge-native leakage statistics.

CRediT authorship contribution statement

Tuan Dao Minh: Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. **Khanh-Trinh Nguyen:** Validation, Data curation, Writing – review & editing. **Duong Nguyen Tien:** Validation, Data curation, Writing – review & editing. **Quoc Khanh Ngo:** Investigation, Resources, Writing – review & editing. **Van-Hau Nguyen:** Validation, Resources. **Le Hoang Son:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics

The exploratory rating pilot scores de-identified, model-generated explanations. It does not recruit learners, collect new educational records, or intervene in instruction. The three raters scored system output already produced from the public XES3G5M release. The authors' institutions do not require research-ethics board review for this form of expert rating of model output.

Data availability

Code and table-to-script artefacts matching this manuscript will be provided to reviewers on request and made public at submission (intended public URL `https://github.com/edu-risk-lab/DH2A-KT`; that address is not yet a live repository). A Zenodo snapshot DOI will be minted at submission. A table-to-script map is in `docs/paper-artifacts.md`. P0's audited edges and reused baselines are vendored under [8]'s release terms (pinned commit `5e3d6a0`, Section 5.2); the companion PDF can be supplied as journal supplementary material while that manuscript is under review. FoundationalASSIST [9] must be obtained from Hugging Face under its Responsible Use Guidelines and cannot be redistributed by the authors.

Acknowledgment

All Tier 1 experiments in this paper are designed to run on a single consumer-grade NVIDIA RTX 3090 (24GB), matching the compute budget already validated by [8] on the same primary benchmarks. FoundationalASSIST [9] was accessed under the Hugging Face dataset's Responsible Use Guidelines for this study.

Appendix A. Graph-only encoder as a forced-reliance diagnostic

The evaluation arm in Table 7 is QKC-T; Zero+Δt is the credited-minimal control. A graph-only diagnostic encoder (chain hyperedges, no question LSTM path; log label v2) exists so a destruction check is interpretable: mean test AUC 0.7516±0.0015 vs. reused GKT 0.8336±0.0010 under P0's GKT budget (L=200; Table A.23). Frozen-weight node-drop p=0.9 passes on all three folds (ΔAUC 0.0323/0.0606/0.0574 on a shared clean AUC per fold); P0 GKT also drops after retraining on a destroyed pairwise graph (0.0877±0.0009). Protocol differs (freeze vs. retrain). Node-drop confirms dependence on some graph input. Degree-preserving rewiring passes on folds 0–2 (ΔAUC 0.0318/0.0341/0.0351; shared clean 0.7485/0.7511/0.7521). Relation-label permutation is near-null on every fold (Table A.24). We do not deploy this encoder.

Table A.23: Diagnostic graph-only track (not QKC-T or Zero+Δt). Test AUC on XES3G5M under P0's matched GKT budget (batch 4, 10 epochs, max seq. len. 200; *not* the clean L=400 protocol of Table 7). DH^2-KT v2 uses path-derived concept-group chain hyperedges and graph-only interaction. Mean±SD over three learner folds. The P0 *simpleKT* column (≈ 0.8747) uses that older leaky protocol and is not comparable to the clean 0.8214±0.0007 in Table 7.

Fold	DH^2-KT v2	GKT (P0)	simpleKT (P0)
0	0.7501	0.8346	0.8744
1	0.7518	0.8338	0.8747
2	0.7531	0.8326	0.8750
Mean±SD	**0.7516±0.0015**	**0.8336±0.0010**	**0.8747±0.0003**
Δ vs. GKT (mean±SD)		−0.0820±0.0025	

Table A.24: Manipulation check on XES3G5M. Destruction is node-drop p=0.9, degree-preserving rewiring, and relation-label permutation on the graph-only diagnostic encoder (log label v2): train on clean hyperedges, re-evaluate frozen weights on destroyed hyperedges (folds 0–2). All three operators now share one clean AUC per fold (`b9_same_ckpt_destruction.json`; max $|\mathrm{AUC}_{\mathrm{local}} - \mathrm{AUC}_{\mathrm{shared}}|$=$2.0\times10^{-9}$). The older printed mismatch (fold 0 0.753 vs. 0.749) was evaluator drift, not an operator effect. Node-drop confirms dependence on some graph input, not correct use of relation structure. Rewiring moves AUC on all three folds (ΔAUC 0.0318–0.0351); label permutation is near-null (expected for a single-kind encoder). GKT: P0 DDR-downstream mean ΔAUC over n=9 fold–seed runs ([8]; retrain on destroyed pairwise graph). simpleKT has no concept-graph input, so graph destruction is undefined. QKC-T and Zero+Δt do not take this check.

Model / destruction	AUC clean	ΔAUC	DDR	Pass?
Diagnostic encoder, node-drop p=0.9 (fold 0)	0.7485	0.0323	0.990	Yes
Diagnostic encoder, node-drop p=0.9 (fold 1)	0.7511	0.0606	0.990	Yes
Diagnostic encoder, node-drop p=0.9 (fold 2)	0.7521	0.0574	0.990	Yes
Diagnostic encoder, degree-preserving rewiring (fold 0)	0.7485	0.0318	0.057	Yes
Diagnostic encoder, degree-preserving rewiring (fold 1)	0.7511	0.0341	0.058	Yes
Diagnostic encoder, degree-preserving rewiring (fold 2)	0.7521	0.0351	0.057	Yes
Diagnostic encoder, relation-label permutation (fold 0)	0.7485	$\approx$0	0.000	No
Diagnostic encoder, relation-label permutation (fold 1)	0.7511	$\approx$0	0.000	No
Diagnostic encoder, relation-label permutation (fold 2)	0.7521	$\approx$0	0.000	No
GKT (P0, mean$\pm$SD), node-drop p=0.9	—	0.0877$\pm$0.0009	0.990	Yes
simpleKT (P0)	—	N/A	—	N/A

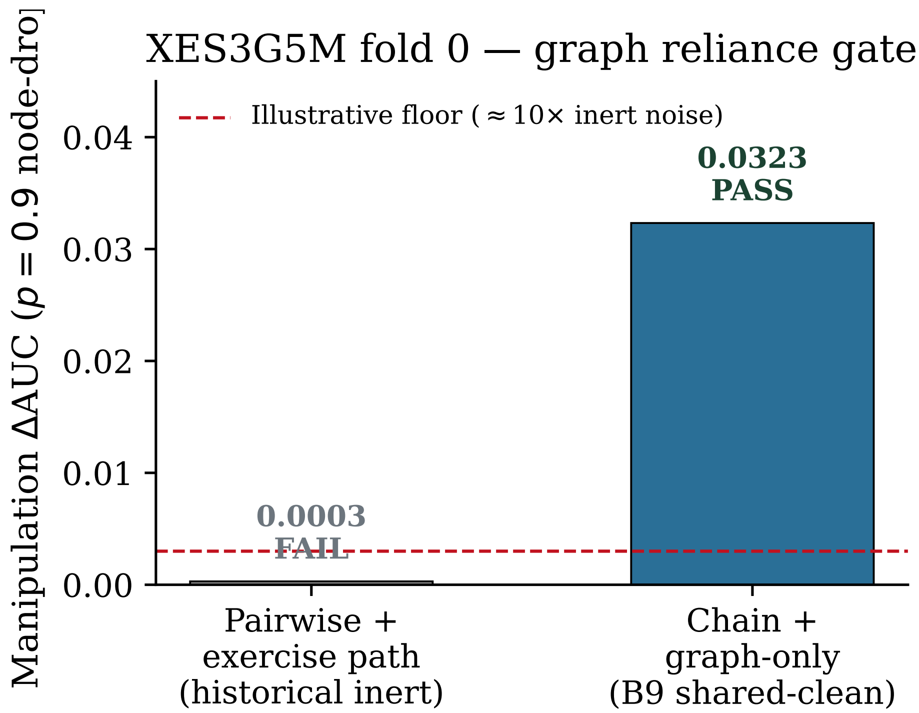

Figure A.3: Manipulation-check ΔAUC on XES3G5M fold 0 under p=0.9 node-drop. The right bar is the B9 shared-clean evaluator (ΔAUC= 0.0323; Table A.24). The left bar is a historical graph-inert path (ΔAUC≈ 0.0003) and is *not* that evaluator. The dashed line at 0.003 is an *illustrative* floor (≈10× the graph-inert noise), not a formal statistical cutoff from [8].

Appendix B. Hyperparameters for Table 7

Table B.25 records, for each model in Table 7, which knobs were searched, which value was used, and how many distinct configurations were tried [30]. We do not claim a matched *tuning* budget.

Table B.25: Training recipes for every model in Table 7, including Zero+Δt (same optimiser as QKC-T; incidence zeroed). "Configs tried" counts distinct hyperparameter vectors, not seeds. DH^2-KT's *architecture* was chosen from a pre-specified ablation ladder (Table 13 plus Q←KC and three Δt injection sites); its *optimiser* recipe is a single vector. GIKT/AKT/*simpleKT*/DKT/DGEKT each used one inherited PYKT vector [27]. AKT and *simpleKT* were not retuned when L moved from P0's 200 to 400. GIKT seed 42 (log `C_gikt_clean_L400_e30b16`) restored best valid AUC at epoch 8 and stopped by epoch 13 under patience 5 (cap 30).

Model	Parameter	Range searched	Selected	How chosen (n configs)
DH^2-KT QKC-T / Zero+Δt	architecture	ablation ladder	LSTM; Zero+Δt zeros incidence	pre-specified gate
DH^2-KT QKC-T / Zero+Δt	batch / epochs / lr / L / patience	—	16 / 30 / 10^{-3} / 400 / 5	one recipe × 5 seeds
GIKT	batch / epochs / patience / L	—	16 / 30 / 5 / 400	match DH^2 training envelope (n=1)
GIKT	remaining pyKT hparams	none	P0/pyKT defaults	inherited (n=1)
AKT	L	{400} override	400	L matched; other hparams inherited (n=1)
AKT	lr / dropout / d_model	none at L=400	P0/pyKT defaults	inherited from L=200 recipe
simpleKT	L / other hparams	{400} override	400 + P0 defaults	inherited (n=1, 5 seeds)
DKT, DGEKT	L / other hparams	{400} override	400 + P0 defaults	inherited (n=1, 5 seeds)

Appendix C. Learner-level bootstrap on one checkpoint pair

Table C.26 records the learner-level paired bootstrap used as a secondary diagnostic in Section 6.2. It resamples 3614 learners that appear on both sides of each contrast; it does not use the 1,093,718-row occurrence join, and it is not a five-seed CI.

Table C.26: Learner-level paired bootstrap on one canonical fold-0 checkpoint pair (1000 resamples of 3614 learners present on both sides). This is a diagnostic of learner sampling, not training-seed variance, and is not the five-seed Zero+Δt gap on Table 7. The contrast is *not* position-aligned: QKC-T is scored on native n=1,093,755 positions; pyKT counterparts on the library intersection n=1,093,720. *simpleKT* AUC 0.8203 is the seed-42 npz, not the five-seed mean 0.8214±0.0007. All three difference intervals exclude zero.

Contrast	AUC QKC-T	AUC other	Δ	95% CI of Δ
vs. GIKT	0.8296	0.8257	+0.0039	[+0.0035, +0.0043]
vs. AKT	0.8296	0.8223	+0.0073	[+0.0068, +0.0077]
vs. *simpleKT*	0.8296	0.8203	+0.0093	[+0.0088, +0.0098]

Appendix D. Observational hint identification check

On FoundationalASSIST we report a standard inverse-propensity (IPW) average treatment effect of $\texttt{hint\_used}_t$ on $\texttt{correct}_{t+1}$, with logistic propensity on pre-t summaries and clipping. Unconfoundedness is assumed, not proven. Same-step `discrete_score` is not the outcome (it is mechanically tied to hint requests). No estimate is reported on XES3G5M (no hint logs). This check is not a contribution.

Using train transitions, the IPW estimate of hint use on *next-step* correctness is −0.207 (naive mean difference −0.255; trimmed IPW −0.211; bootstrap 95% CI [−0.230, −0.187] on 50k-row resamples; n_{treated}=36,151). Propensity scores hit the clip floor ([0.05, 0.219]), indicating weak overlap. The negative sign reflects that weaker learners request more hints; it must *not* be read as the effect of providing hints, and it is not a recommendation against hints. We did not compute Rosenbaum bounds or an E-value.

Table D.27: FoundationalASSIST fold 0 IPW identification check of $\texttt{hint\_used}_t$ on $\texttt{correct}_{t+1}$ under pre-t confounders. Propensity mass hits the clip floor. The negative IPW estimate reflects that weaker learners request more hints; it is *not* the effect of providing hints.

Quantity	Value
Hint ATE (IPW) / naive mean diff	−0.207 / −0.255
trimmed IPW (propensity 10–90%)	−0.211
bootstrap 95% CI (subsample)	[−0.230, −0.187]
n treated / control (train transitions)	36,151 / 660,321
propensity $[\min, \max]$	[0.05, 0.219]

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Cursor (Composer) in order to draft and revise LaTeX wording, tables, and cover-letter text under author direction. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The Tier 2 Diagnostician/Critic/Tutor agents (Qwen2.5-7B via Ollama) are experimental components of the reported system, not manuscript-writing tools, and are described in Section 6.4.

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